Unit 7: Final Engineering Design Challenge

Quiz 7A: Capstone Planning, Problem Definition, and Project Management

Name: ____________________ **Period:** ____________________

Date: ____________________ **Score:** ____________________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. The first major step in the final design challenge should be...
   A. Define the problem and mission need
   B. Start building immediately
   C. Choose colors
   D. Make a final slide only

2. Project management helps teams...
   A. Plan tasks, deadlines, roles, and progress
   B. Avoid documentation
   C. Skip testing
   D. Use tools without permission

3. A Gantt chart or task board can help track...
   A. Schedule and responsibilities
   B. Only grades
   C. Only sketches
   D. Only material color

4. Why should teams define criteria and constraints again in Unit 7?
   A. The final project has its own mission and limits
   B. Criteria no longer matter
   C. It only helps the teacher
   D. It replaces research

5. A milestone is...
   A. A checkpoint or important progress target
   B. A random part
   C. A safety rule only
   D. An unused file

6. A risk in project planning is...
   A. Something that could prevent success or cause delay/failure
   B. A guaranteed success
   C. Only a rubric score
   D. Always beneficial

7. Which is a strong team role practice?
   A. Assigning responsibilities and checking progress
   B. Everyone assumes someone else will do it
   C. No roles
   D. Only one person works

8. Why use a design notebook during the capstone?
   A. To document decisions, evidence, tests, and revisions
   B. To avoid the portfolio
   C. To hide failures
   D. To replace all communication

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

_____	**Capstone**	A. A final project that combines many course skills.
_____	**Project management**	B. Planning and tracking tasks, roles, schedule, and progress.
_____	**Milestone**	C. An important checkpoint or deadline.
_____	**Risk**	D. A possible issue that could affect success.
_____	**Constraint**	E. A limit or requirement the design must follow.

Part C: Short Response

1. Explain why project management becomes more important during the final challenge.

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2. Identify one possible project risk and how a team could reduce it.

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Unit 7: Final Engineering Design Challenge

Quiz 7B: Testing, Data Collection, Iteration, and Design Changes

Name: ______________________ **Period:** ______________________

Date: ______________________ **Score:** ______________________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. Prototype testing should be connected to...
   A. Criteria and performance goals
   B. Only team preference
   C. Only decoration
   D. No documentation

2. Data collection is useful because...
   A. It provides evidence for design decisions
   B. It replaces all sketches
   C. It guarantees success
   D. It hides failures

3. Iteration means...
   A. Improving the design based on feedback or test results
   B. Never changing anything
   C. Starting over randomly
   D. Only presenting

4. A design change log records...
   A. What changed, why it changed, and when
   B. Only team names
   C. Only photos
   D. Only final scores

5. Why test one change at a time when possible?
   A. To understand which change affected performance
   B. To make testing confusing
   C. To avoid evidence
   D. To use more materials

6. A failed prototype test should lead to...
   A. Failure analysis and redesign planning
   B. Hiding the result
   C. Stopping the project
   D. Blaming a teammate

7. Which is a clear test metric?
   A. Payload held for 10 seconds at 500 g
   B. It seems good
   C. Nice design
   D. Cool result

8. Why repeat tests?
   A. To check consistency and reliability
   B. To make results less clear
   C. To skip analysis
   D. To copy another team

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

_____ **Prototype**	A. A testable model or version of a design.
_____ **Test metric**	B. A measurable performance target or result.
_____ **Iteration**	C. A cycle of testing and improving a design.
_____ **Change log**	D. A record of design changes and reasons.
_____ **Reliability**	E. The ability to perform consistently over repeated tests.

Part C: Short Response

1. Describe how a team should respond to a failed prototype test.

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2. Explain why a change log is useful during a long design project.

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Unit 7: Final Engineering Design Challenge

Quiz 7C: Final Portfolio, Design Review, and Engineering Defense

Name: ____________________ **Period:** ____________________

Date: ____________________ **Score:** ____________________

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. A final engineering portfolio should show...
   A. The full design process, evidence, testing, and final documentation
   B. Only final photos
   C. Only grades
   D. Only the best sketch

2. An engineering defense requires students to...
   A. Justify choices using evidence
   B. Avoid questions
   C. Hide weaknesses
   D. Only read slides

3. Which evidence strengthens a final design review?
   A. Test data, CAD images, drawings, sketches, and analysis
   B. Only opinions
   C. Only team names
   D. Only jokes

4. Why include limitations in the final review?
   A. To show what still needs testing or improvement
   B. To make the design fail
   C. To lower the score automatically
   D. To avoid questions

5. A strong final presentation should connect back to...
   A. Problem statement, criteria, constraints, and test results
   B. Only the title slide
   C. Only the logo
   D. Only the final day

6. What should teams explain about redesign?
   A. What changed and what evidence caused the change
   B. Only that they changed something
   C. Nothing
   D. Only the color

7. Which question is likely in an engineering defense?

A. What evidence supports your design choice?
B. What is your favorite snack?
C. Can you skip testing?
D. Why no documentation?

8. The final portfolio is stronger when it includes...
   A. Process evidence as well as final product evidence
   B. Only polished results
   C. No failures
   D. No sketches

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

____	**Portfolio**	A. A collection of documents showing the full design process and final work.
____	**Engineering defense**	B. A question-and-answer explanation of design decisions using evidence.
____	**Evidence**	C. Data, sketches, models, drawings, or observations supporting a claim.
____	**Limitation**	D. A weakness, risk, or unresolved issue in a design.
____	**Final design review**	E. A formal presentation of final design process, product, evidence, and next steps.

Part C: Short Response

1. Explain why a final portfolio should include failed tests or redesign evidence.

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2. Write one engineering defense question a team should be ready to answer.

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